ADITYA REDDY

Software Engineer 2

+1 (206) 571 1923 | adityakasar2910@gmail.com | LinkedIn | GitHub

SUMMARY

Software Engineer with **4+ years of experience** designing and developing **distributed backend and full-stack systems** across cloud-native environments. Strong experience in **Java, Spring Boot, Python, React, and TypeScript**, with hands-on ownership across **microservices architecture, data systems, APIs, CI/CD, observability, security, and production reliability**. Experienced in building and operating solutions on **AWS and Azure**, along with developing **GenAI and RAG-based applications** for engineering and operational use cases.

EXPERIENCE

Software Development Engineer 2 | **Amazon Web Services – AWS FinTech**, *Seattle, WA* **Apr 2025 – Present**

- Designed and implemented **Java 21 microservices** for EC2 deal modeling across **Lambda, ECS, DynamoDB, and RDS**, separating calculation, workflow, and persistence layers for enterprise **pricing and P&L analysis**.
- Engineered an **EC2 snapshot pipeline** using **AWS Glue and Step Functions** to extract, validate, and persist **10+ pricing, cost, and usage datasets** across **RDS, DynamoDB, and S3**, ensuring consistent inputs for reproducible P&L calculations.
- Built a **React/TypeScript spreadsheet-style grid** capable of handling **100+ SKUs**, with **useReducer-driven state management**, memoized rendering, autosave, and optimistic concurrency for scalable, responsive modeling.
- Enhanced a shared **React/TypeScript data-grid library** built on **TanStack** with **Cloudscape** components, adding virtualization, editing, filtering, and keyboard navigation while resolving feature gaps and integration issues for **8+ teams**.
- Engineered reusable **cross-account DNS and API networking infrastructure**, cutting endpoint provisioning time by **2 weeks** and standardizing network setup across deployed applications.
- Improved operational visibility across **176 production APIs** by implementing CloudWatch metrics, dashboards, and automated alarms while coordinating cross-team UAT, release readiness, and production rollout activities.
- Built **AI assisted on call workflows** to triage failures across **APIs, databases, deployments, and performance signals**, helping investigate **30+ production issues** and drive **20+ fixes** through resolution while reducing repeat incidents.
- Developed an **agentic AI engineering assistant** using **AWS Bedrock, Claude, Strands, MCP, and RAG** to retrieve **code, design documents, runbooks, and operational context** for developers and on call engineers, reducing **debugging and ramp up time by 60%**.

Software Engineer | **Bank of America**, *Remote* **Feb 2024 – Mar 2025**

- Owned design and delivery of **Java/Spring Boot payment-routing services** on **Azure AKS**, processing **100K+ authorizations/day** while maintaining a **99.9% availability SLO**.
- Secured payment workflows with **Kafka field-level encryption, Azure Key Vault, and least-privilege access**, coordinating rollout across **5+ services and 8 event topics** with under **5 ms P95 encryption overhead**.
- Redesigned data access across **Oracle, Cassandra, and Redis**, separating transactional, historical, and cached workloads and cutting **P95 API latency from 420 ms to 260 ms**.
- Built a **RAG incident-intelligence service** with **Azure AI Search, Azure OpenAI, and FastAPI**, surfacing relevant runbooks and postmortems for on-call engineers and contributing to faster incident resolution.
- Strengthened release and production operations with **GitHub Actions, ArgoCD, Azure Monitor, Application Insights, and automated testing**, expanding backend coverage to **90%** and adding deployment gates for critical services.

Graduate Research Assistant | **Computer Science Dept., Illinois State University**, *Normal, IL* **Jul 2023 – Jan 2024**

- **Designed and architected** a full-stack predictive maintenance platform spanning a hybrid Python/C++ data pipeline, LSTM-based ML models, and a React monitoring dashboard to deliver end-to-end IoT sensor intelligence at scale.
- Built a **Python/C++ pipeline** to process **500GB+** of monthly IoT sensor data, leveraging Python for LSTM training and C++ for high-throughput preprocessing and inference.
- Trained **LSTM time-series models** on sensor and GPS data to detect maintenance risk patterns, reducing manual inspection effort by **30%**.
- Engineered a **real-time data processing pipeline** for cleaning, normalization, aggregation, and feature engineering, reducing end-to-end prediction latency by **70%**.

Software Engineer | **Client – Walmart**, *Hyderabad, India* **Aug 2020 – Jul 2022**

- Developed **Java/Spring Boot services** supporting **10K+ orders/day**, coordinating inventory, payments, and fulfillment across distributed client systems and supporting reliable end-to-end order processing.
- Shipped 30+ reusable, accessible UI components across several web apps in close collaboration with PMs and UX Designers, and engineered 15+ scalable API endpoints and database integrations in C# and Java, improving feature delivery speed and system reliability
- Led an **Angular-to-React migration**, improving development velocity by **40%** and reducing technical debt by **60%** with zero-downtime transition.
- Built and consumed RESTful APIs secured with Spring Security, ensuring authentication, authoriza tion, and data protection across services.

PROJECTS

LLM Multi Engine | *NGINX, CUDA, Python, LangChain, LangGraph, LoRA*

- Integrated **Punica's Segmented Gather Matrix-Vector (SGMV) CUDA kernels** for multi-tenant LLM serving, increasing throughput by **12x** while concurrently serving **LoRA models**.

- Optimized **GPU inference pipelines and CUDA kernel execution** for low-latency, high-throughput inference, supporting scalable **real-time LLM serving**.
- Architected **agentic AI workflows using LangChain and LangGraph** to orchestrate multi-step reasoning and connect GPU inference pipelines with **external APIs and data sources**.

FinSight AI | *AWS-native RAG Platform for Payments Compliance*

- **High-Throughput Inference:** Served **200 req/sec** at **sub-500ms latency** by fine-tuning **LLaMA-3.1** with **QLoRA** on **Amazon SageMaker** and deploying inference via **vLLM**, using retrieval grounding for answer quality and fine-tuning for output consistency.
- **End-to-End RAG Platform:** Enabled **KYC/AML** and payment compliance analyst search via a **Python/FastAPI** pipeline on **LlamaIndex** and **Amazon OpenSearch Serverless**, with **S3** ingestion and **DynamoDB** merchant telemetry.

PeerVault | *P2P Encrypted File Distribution*

- **Bandwidth-Efficient Protocol:** Reduced network bandwidth by ∼**50%** over JSON-over-HTTP by building a decentralized file-sharing system in **Java** with **RSA/AES hybrid encryption**, a **Kademlia DHT** for peer discovery, and a custom **Protocol Buffers** binary protocol.
- **I/O Throughput Optimization:** Scaled cluster throughput from **1.2 Gbps** → **3.6 Gbps** on a **10-node cluster** while lowering CPU utilization under sustained load by integrating **Linux io_uring** via **JNI** for asynchronous disk and network I/O.

TECHNICAL SKILLS

Languages: Java, Python, C++, TypeScript, JavaScript, CUDA
Backend & APIs: Spring Boot, REST APIs, GraphQL, Microservices
Frontend: React.js, Redux, Angular, Cloudscape, TanStack Table
Data, ML & GenAI: Kafka, Apache Airflow, AWS Glue, LSTM, Time-Series Modeling, Amazon Bedrock, Claude, MCP, LLMs, LangChain, LangGraph, Agentic AI Workflows, Multi-Step Reasoning
Cloud, Databases & Infrastructure: AWS, Microsoft Azure, PostgreSQL, Oracle, DynamoDB, Redis, Terraform, Docker, Kubernetes, Jenkins, CI/CD, NGINX
Security, Testing & Observability: Spring Security, OAuth 2.0, IAM, JUnit, Mockito, Cypress, CloudWatch, Azure Monitor

EDUCATION

Master's in Computer Science (GPA: 3.7) **Aug 2022 – May 2024**
Illinois State University *Normal, IL*